

Absolutely. I would actually design this as if it were a real SaaS product that could scale to hundreds of thousands of users.

I **would not** keep SQLite for production. My recommendation is:

- **Development:** SQLite + Prisma
- **Production:** PostgreSQL + Prisma (Neon/Supabase/Railway)

Since Prisma supports both, you can start with SQLite locally and migrate later with almost no code changes.

Product Requirements Document (PRD)

Project Name

FrameWise AI

AI-powered PC Gaming Performance Analyzer

1. Overview

FrameWise AI is a web platform that automatically detects a user's PC hardware, predicts expected FPS across thousands of games, analyzes hardware bottlenecks, recommends upgrades based on the user's budget and country, and provides AI-generated optimization advice.

Unlike traditional "Can You Run It?" websites, FrameWise focuses on:

- Expected FPS
- Performance score
- Upgrade recommendations
- Local market pricing
- AI explanations

Goals

Users should be able to answer these questions within 30 seconds:

- Can my PC run this game?
- How many FPS should I expect?
- Which component is limiting performance?
- What should I upgrade first?
- How much will that upgrade cost in my country?

Target Audience

- PC gamers
- Laptop gamers
- PC builders
- Students
- Gaming cafés
- Hardware reviewers

Tech Stack

Frontend

- Next.js 15 App Router
- React 19
- TypeScript
- Tailwind CSS
- shadcn/ui
- React Hook Form

- Zod
- TanStack Query
- Framer Motion
- Recharts

Backend

- Next.js Route Handlers
- Prisma ORM
- SQLite (Development)
- PostgreSQL (Production)
- NextAuth/Auth.js
- JWT Sessions

Storage

- Cloudinary (game covers if needed)

Caching

- Redis

AI

- OpenRouter API

Deployment

- Vercel
- Neon PostgreSQL

Hardware Detection

Because browsers cannot access detailed hardware information (CPU/GPU model, RAM capacity, etc.) for security reasons, true automatic detection requires a **small companion desktop application** (Windows initially) that securely sends the hardware profile to the

authenticated user. The web app can detect only limited information (screen resolution, browser, OS hints, etc.) on its own.

Companion App Responsibilities

- Detect CPU, GPU, RAM, storage, motherboard, monitor resolution, OS.
- Authenticate with the user's account.
- Upload hardware profile.
- Periodically sync if hardware changes.

User Flow

Landing



Register



Login



Install Hardware Detector



Hardware Synced



Dashboard



Browse Games



Game Details



AI Upgrade Suggestions

Authentication

Public

- Register
- Login
- Forgot Password
- Reset Password
- Verify Email

Protected

- Dashboard
- Games
- Game Details
- Upgrade Planner
- AI Assistant
- Profile
- Settings

User Roles

User

Admin

Public Pages

/

About

Features

Pricing

Games

Game Details (public read-only)

/login

/register

/forgot-password

/reset-password

/privacy

/terms

/contact

/blog

/changelog

Protected Pages

/dashboard

/profile

/settings

/my-pc

/upgrade-planner

/ai

/history

/account

/logout

Admin Pages

/admin

/admin/games

/admin/game/new

/admin/game/edit

/admin/hardware

/admin/prices

/admin/users

/admin/benchmarks

/admin/blog

Dashboard

Sections

Hardware Summary

Gaming Score

Detected Hardware

Performance Overview

Latest Games

Upgrade Recommendation

Recent Searches

Quick Search

Games Page

Search

Filters

Genre

Developer

Publisher

Release Year

Minimum FPS

Resolution

Sorting

Newest

Highest FPS

Lowest FPS

Most Popular

Game Details

Hero

Cover

Title

Developer

Publisher

Genre

Release Date

Engine

Performance

720P

1080P

1440P

4K

Ultra

High

Medium

Low

FPS Color

Requirements

Minimum

Recommended

AI Analysis

Performance Explanation

Upgrade Advice

Optimization Tips

Budget Recommendations

Related Games

Upgrade Planner

Input

Budget

Preferred Resolution

Preferred FPS

Graphics Quality

Output

Recommended GPU

Recommended CPU

Estimated FPS

Price

Country

Store

AI Assistant

Powered by OpenRouter

User asks

Can I play GTA VI?

How can I reach 144 FPS?

What should I upgrade first?

My budget is \$300.

AI returns

Hardware analysis

Upgrade advice

Optimization guide

Expected FPS

Settings

Theme

Country

Currency

Preferred Resolution

Language

Notification Preferences

Profile

- Avatar
- Username
- Email
- Hardware
- Statistics

Search

- Global Search
- Games
- GPU
- CPU
- Developers
- Publishers

Performance Indicators

FPS	Label	Color
0-29	Poor	Red
30-59	Playable	Yellow
60-89	Smooth	Green
90-119	High Refresh	Blue
120+	Competitive	Purple

Database Modules

Users

Games

Developers

Publishers

Genres

Hardware

Benchmarks

Requirements

Recommendations

Prices

Countries

Currencies

AI Conversations

Search History

Folder Structure

framewise-ai/

```
|— app/  
|   |— (marketing)/
```

```
|
|
|   |— page.tsx
|   |— pricing/
|   |— features/
|   |— about/
|   |— contact/
|   |— games/
|
|   |— (auth)/
|   |   |— login/
|   |   |— register/
|   |   |— forgot-password/
|   |   |— reset-password/
|
|   |— (dashboard)/
|   |   |— dashboard/
|   |   |— my-pc/
|   |   |— games/
|   |   |— games/[slug]/
|   |   |— upgrade-planner/
|   |   |— ai/
|   |   |— profile/
|   |   |— settings/
|   |   |— history/
|
|   |— admin/
|
|   |— api/
|
|   |— globals.css
|   |— layout.tsx
|
|— components/
|   |— ui/
|   |— game/
|   |— hardware/
|   |— charts/
|   |— dashboard/
|   |— ai/
|   |— forms/
```

```
|
|   ├── shared/
|   └── layout/
|
|── features/
|   ├── auth/
|   ├── games/
|   ├── hardware/
|   ├── benchmarks/
|   ├── ai/
|   ├── profile/
|   ├── settings/
|   └── upgrade/
|
|── actions/
|
|── hooks/
|
|── services/
|   ├── openrouter/
|   ├── hardware/
|   ├── benchmarks/
|   ├── pricing/
|   └── cache/
|
|── lib/
|   ├── prisma.ts
|   ├── auth.ts
|   ├── redis.ts
|   ├── utils.ts
|   ├── validators/
|   └── constants/
|
|── prisma/
|   ├── schema.prisma
|   ├── migrations/
|   └── seed.ts
|
|── types/
|── config/
```

```
|— public/  
|— middleware.ts  
└— package.json
```

API Routes

```
POST  /api/auth/login  
POST  /api/auth/register  
POST  /api/auth/logout
```

```
GET    /api/user
```

```
GET    /api/hardware
```

```
POST    /api/hardware/sync
```

```
GET    /api/games
```

```
GET    /api/games/:slug
```

```
GET    /api/search
```

```
POST    /api/upgrade
```

```
POST    /api/ai/chat
```

```
GET    /api/profile
```

```
PATCH  /api/profile
```

```
GET    /api/settings
```

```
PATCH  /api/settings
```

External Integrations

Authentication

- Auth.js (NextAuth)
- Google OAuth
- GitHub OAuth
- Email/Password

AI

- OpenRouter API
- GPT-5.5 / Claude Sonnet / Gemini models (user-selectable if desired)

Game Data

- IGDB API (game metadata)
- RAWG API (fallback for covers/screenshots)

Local Pricing

Rather than scraping stores directly, maintain a normalized `HardwarePrice` table updated by scheduled jobs or approved data sources. This lets recommendations remain fast and consistent while supporting country-specific pricing (EGP, SAR, USD, etc.).

Email

- Resend

Analytics

- PostHog or Umami
- Google Analytics (optional)

Error Monitoring

- Sentry

Suggested AI Features

Instead of a simple chatbot, give the AI structured capabilities:

- Explain why a game performs poorly on the user's PC.
- Recommend the best upgrade within a specified budget.
- Compare two GPUs or CPUs in the context of the user's hardware.
- Suggest optimal in-game graphics settings to reach a target FPS.
- Recommend games that are well suited to the user's current hardware.
- Answer hardware-related questions using the user's synced PC profile as context.